Who Bears the Congestion Price?

— Fare Burden and Trip Pattern Shifts in NYC Taxi/FHV Trips

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Policy Background

Starting January 2025, vehicles entering the CBD congestion zone face a fee.

Private cars

9$/day

Yellow Taxi

0.75$/trip

Uber / Lyft

1.5$/trip

Project Questions

1 Who bears the highest relative fee burden?

- Manhattan trips v.s. outer-trips?
- Different zones within Manhattan?
- Airport trips v.s. non-airport trips?

2 Did the fee reduce trips in taxi and Uber/Lyft?

A causal inference problem to find the relationship between volume change and fee-burden

Data Structure

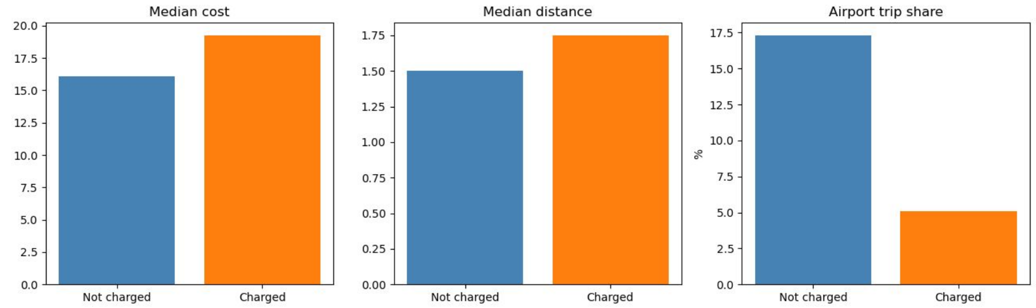
Resource: public TLC trip data

Data type: Yellow Taxi and Uber / Lyft.
Each trip records geographic info (PU/DO location, distance, etc.) , time stamps (PU/DO time, duration), financial info (base fare, tolls, congestion fee, tips paid).

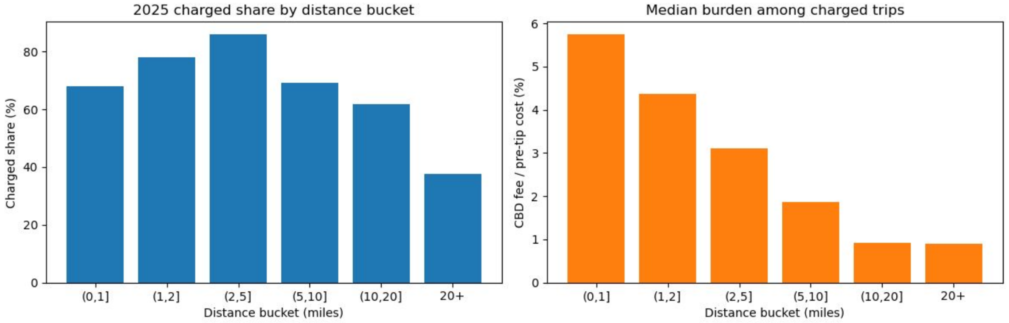
Time window: Feb-Jun 2024 (before policy) vs. Feb-Jun 2025 (after policy).
Same-month comparison excludes seasonal effects; January 2025 excluded as the policy's transition month.

EDA Findings

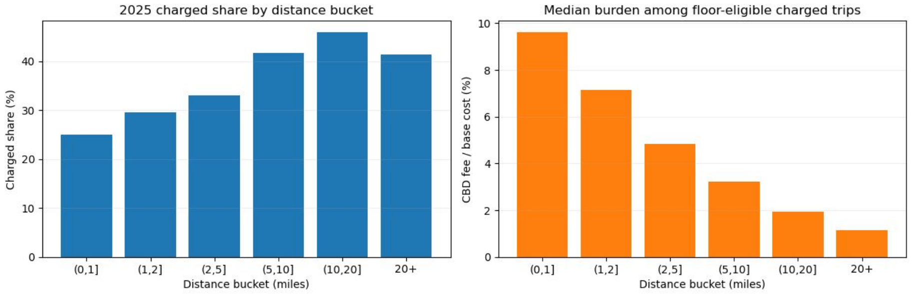
Charged Yellow trips differ from uncharged trips



Yellow: short trips carry higher relative burden



HVFHV: distance changes both exposure and burden



EDA takeaway: burden is uneven, so the analysis first maps where the fee is largest relative to trip cost.

Measurements

$$\text{fee burden} = \frac{\text{CBD fee}}{\text{pretip cost} - \text{CBD fee}}$$

zone disruption score

= **average fee burden over charged trips in zone x direction**

Metric for Fee Burden Ranking

zone x Pickup side

and

zone x Dropoff side

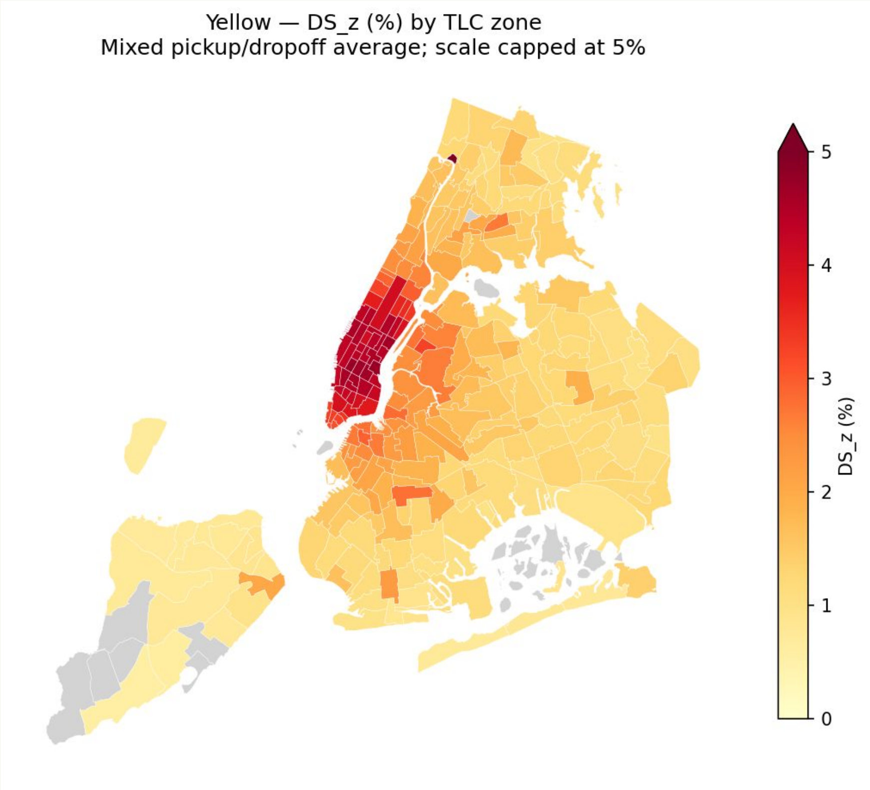
Shared Unit of Analysis

change in trip volume

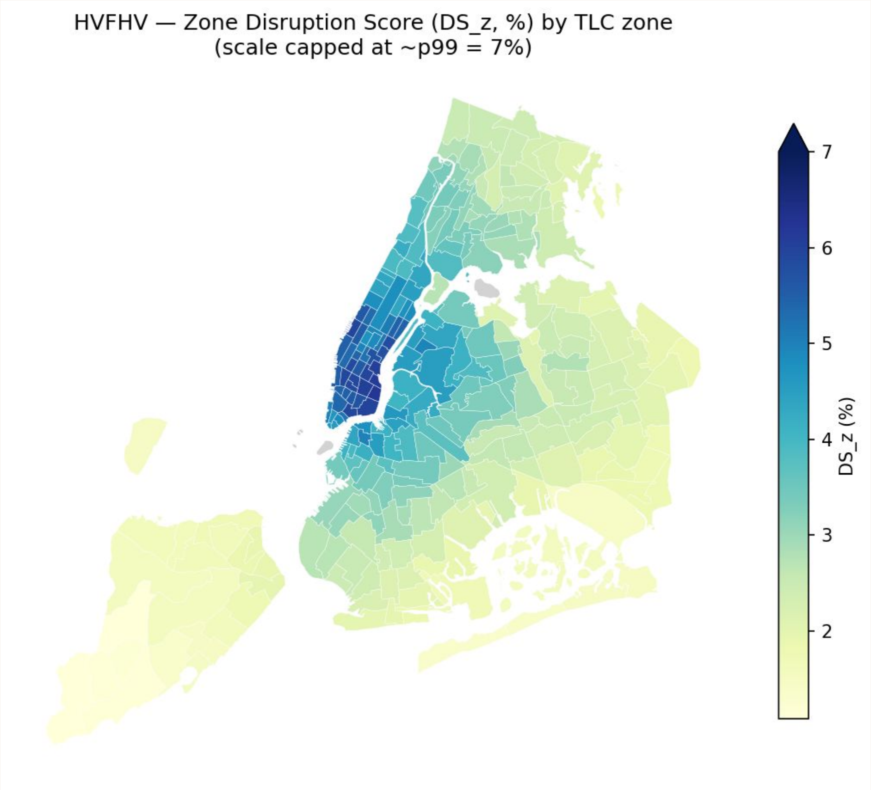
Outcome for inference

Findings: Burden Analysis

Highest relative burden concentrates in short, lower-cost Manhattan trips.



Yellow Taxi



Uber / Lyft

Rankings stay stable across denominator checks — the ranking does not depend on how we do the math.

Modeling Ladder for Inference

Regression with controls

	It asks	Main comparison	Reads as
Model 1	Do high-burden zones lose more trips?	high- vs low-burden zones	descriptive only
Model 2	Did higher-exposure zones change more?	more- vs less-exposed, over time	DiD — needs parallel trends
Model 3	Did the higher-fee service change more?	Uber/Lyft vs taxi, same zone	cross-service — strong assumptions

Why regression, not a fancier model: a better fit cannot fix an unfair comparison

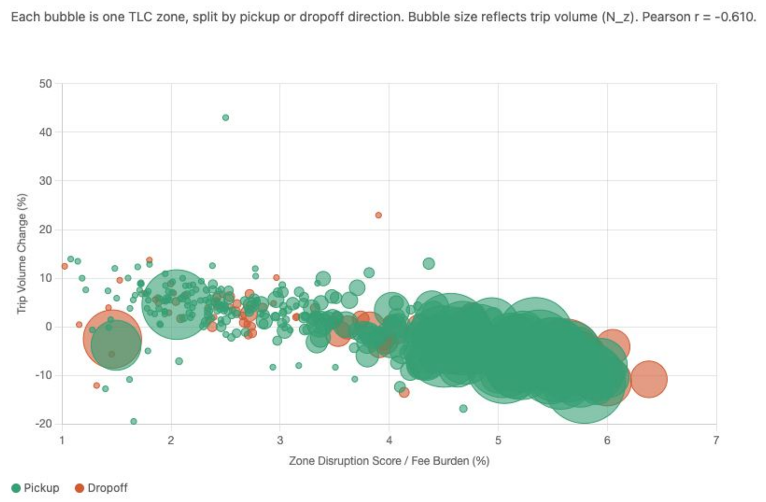
Findings: Volume Models

Negative estimates appear, but the checks do not support a clean causal volume claim.

Model checks

Model	Comparison	Estimate	Diagnostic
Model 1	burden vs volume	r = -0.61 / -0.17	moves with controls
Model 2	exposure panel	β ≈ -0.12 → -11%	2023-24 placebo ≈ -26%
Model 3	Uber/Lyft vs taxi	β ≈ -0.06 → -5.9%	same-fee split differs

Volume pattern is visible



The chart motivates the volume question; the table shows why we report association, not proof.

Limited Causal Interpretation

TLC records show completed trips, not every travel choice.

Policy operates through a wider travel system

Congestion fee → Rider choices → Completed TLC trips

Data limitation

We observe completed TLC trips, not the full set of choices riders considered.

Unobserved channels

private cars | transit | walking/biking
app searches | canceled or skipped trips

Comparison issue

High-exposure zones are not random: they differ in geography, trip length, costs, and demand trends.

So weaker volume may reflect the fee, but also geography, platform competition, or mode substitution.

Where This Goes Next

Each next step targets one limitation in the current evidence.

Sharpen the comparison

Use longer pre-policy panels, placebo windows, and boundary or matched-zone checks.

Look beyond TLC trips

Add transit, bike, vehicle-entry, and speed data to see what happens outside completed taxi and Uber/Lyft trips.

Separate fee effects from platform dynamics

Add provider-specific baselines and rider-neighborhood context to separate fee response from Uber/Lyft market shifts.

These extensions would improve interpretation, but each one needs its own design.